Logistic Chain Equilibrium on Spherical Port Networks

An Entropic Optimal Transport Approach

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Abstract

We develop a unified mathematical framework for the equilibrium analysis of global maritime logistics on a directed, weighted graph $G = (V, E)$ embedded on the two-sphere $\mathbb{S}^2$. Vertices represent ports weighted by *net tonnage* (imports minus exports), and edges carry great-circle distances weighted by *net shipping distance*. A *logistic chain* couples these two weights multiplicatively, defining a measure of transport effort on each route. We model routing decisions as a multinomial logistic (softmax) regression over outgoing arc distances, which we show is equivalent to entropy-regularised optimal transport (EOT) with a Schoenberg-positive kernel on $\mathbb{S}^2$. Using this equivalence we prove existence, uniqueness, and characterise the equilibrium flow as a doubly-stochastic Sinkhorn–Wilson fixed point. We give a provably convergent iterative algorithm, analyse its linear convergence rate in terms of the spectral gap of the transport kernel, and illustrate the theory with worked examples and tables.

The paper ends with "The End"

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1 Introduction

Global maritime trade moves more than 80% of world cargo by volume [16]. Understanding how equilibrium emerges in such networks—where thousands of vessels simultaneously optimise transit costs—is of fundamental importance to logistics planning, carbon accounting, and port infrastructure investment.

Classical network-flow models [1] enforce conservation but are silent on *how* routes are chosen. The gravity model of trade [15, 2] captures bilateral flows statistically but lacks a dynamic routing rule. Entropy-regularised optimal transport (EOT) [4, 17] bridges these gaps: it yields a smooth, differentiable flow that satisfies supply/demand constraints while penalising long-haul routing.

Contributions.

C1. We formalise the spherical port network $G = (V, E)$ with tonnage and distance weights, and define the *logistic chain* as the natural coupling of these weights (Section 3).

C2. We derive the logistic routing rule as maximum-entropy inference and identify it with the softmax of the distance disutility (Section 4).

C3. We formulate the global equilibrium as a convex programme and prove uniqueness via strict convexity of the Kullback–Leibler divergence (Section 5).

C4. We characterise the closed-form solution using Sinkhorn potentials and establish geometric convergence of the iterative algorithm (Section 6).

C5. We analyse the role of the spherical geometry—specifically Schoenberg's theorem—in guaranteeing a positive-definite kernel (Section 7).

2 Mathematical Preliminaries

2.1 Directed Graphs and Incidence Matrices

Let $G = (V, E)$ be a finite directed graph with $n = |V|$ vertices and $m = |E|$ directed edges. The *signed incidence matrix* $\mathbf{B} \in \mathbb{R}^{n\times m}$ has

$$B_{v,e} = \begin{cases} +1 & \text{if edge } e \text{ enters } v, \\ -1 & \text{if edge } e \text{ leaves } v, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

The (unnormalised) *graph Laplacian* is $L = \mathbf{B}\mathbf{B}^\top \in \mathbb{R}^{n\times n}$. For strongly connected G, $\operatorname{rank}(L) = n - 1$.

2.2 Optimal Transport on Discrete Spaces

Let $\boldsymbol{r}, \boldsymbol{c} \in \mathbb{R}^n_{>0}$ with $\mathbf{1}^\top \boldsymbol{r} = \mathbf{1}^\top \boldsymbol{c} = 1$ be discrete probability vectors. The *Kantorovich optimal transport* problem [17] is

$$\mathrm{OT}(\boldsymbol{r}, \boldsymbol{c}) = \min_{\Pi \in \mathcal{U}(\boldsymbol{r},\boldsymbol{c})} \langle C, \Pi \rangle, \tag{2}$$

where $C \in \mathbb{R}^{n\times n}_{\geq 0}$ is the cost matrix, $\langle \cdot, \cdot \rangle$ is the Frobenius inner product, and

$$\mathcal{U}(\boldsymbol{r}, \boldsymbol{c}) = \left\{ \Pi \in \mathbb{R}^{n\times n}_{\geq 0} : \ \Pi \mathbf{1} = \boldsymbol{r}, \ \Pi^\top \mathbf{1} = \boldsymbol{c} \right\} \tag{3}$$

is the transport polytope.

2.3 Entropy Regularisation and the Sinkhorn Kernel

The *entropy-regularised* variant [4] adds a Kullback–Leibler penalty:

$$\mathrm{OT}_\varepsilon(\boldsymbol{r}, \boldsymbol{c}) = \min_{\Pi \in \mathcal{U}(\boldsymbol{r},\boldsymbol{c})} \langle C, \Pi \rangle + \varepsilon \ \mathrm{KL}(\Pi \| \boldsymbol{r}\boldsymbol{c}^\top), \tag{4}$$

where $\mathrm{KL}(\Pi \| \boldsymbol{r}\boldsymbol{c}^\top) = \sum_{ij} \Pi_{ij} \ln \frac{\Pi_{ij}}{r_i c_j}$. The unique minimiser takes the form

$$\Pi^*_{ij} = u_i \, K_{ij} \, v_j, \qquad K_{ij} = \exp\left(-\frac{C_{ij}}{\varepsilon} \right), \tag{5}$$

for scaling vectors $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n_{>0}$ found by the *Sinkhorn–Knopp* iterations [13].

2.4 Differential Geometry of the Two-Sphere

Let ports be located at unit vectors $\hat{p}_i \in \mathbb{S}^2 \subset \mathbb{R}^3$. The *great-circle distance* between ports A and B on a sphere of radius r is

$$d(A, B) = r \arccos\!\big(\hat{A} \cdot \hat{B}\big), \qquad \hat{A}, \hat{B} \in \mathbb{S}^2. \tag{6}$$

Theorem 2.1 (Schoenberg, 1942 [12])**.** *A continuous function $\psi : [0, \pi] \to \mathbb{R}$ gives rise to a positive-definite kernel $K(\hat{A}, \hat{B}) = \psi(\theta_{AB})$ on $\mathbb{S}^2$ if and only if $\psi(\theta) = \sum_{k=0}^{\infty} a_k P_k(\cos\theta)$ with $a_k \geq 0$ for all k, where P_k are Legendre polynomials. In particular, $K(\hat{A}, \hat{B}) = \exp(-\lambda d(A, B))$ is positive-definite on $\mathbb{S}^2$ for all $\lambda > 0$.*

3 The Spherical Port Network Model

3.1 Network Definition

Definition 3.1 (Spherical Port Network)**.** A *spherical port network* is a quadruple $\mathcal{N} = (G, \hat{\cdot}, w, d)$ where:

- $G = (V, E)$ is a strongly connected directed graph with $V = \{p_1, \ldots, p_n\}$ and $E \subseteq V \times V$;
- $\hat{p}_i \in \mathbb{S}^2$ is the geographic location of port p_i;
- $w : V \to \mathbb{R}$ is the *net tonnage weight*, $w(v) = T_{\text{in}}(v) - T_{\text{out}}(v)$;
- $d : E \to \mathbb{R}_{>0}$ is the great-circle distance (6) on each arc.

Definition 3.2 (Supply, Demand, and Balance)**.** Define the *export supply* $s_v = \max(0, -w(v))$ and *import demand* $r_v = \max(0, w(v))$ at each port. The *global balance* condition requires

$$\sum_{v \in V} w(v) = 0 \quad \Longleftrightarrow \quad \sum_v s_v = \sum_v r_v =: \mathcal{T}, \tag{7}$$

where $\mathcal{T}$ is the total throughput of the network.

3.2 The Logistic Chain

Definition 3.3 (Logistic Chain)**.** Given a flow $f : E \to \mathbb{R}_{\geq 0}$, the *logistic chain value* of arc $(A \to B) \in E$ is

$$\mathcal{L}(A \to B;\, f) = f(A \to B) \cdot d(A, B). \tag{8}$$

The *total network logistic cost* is

$$\mathcal{C}(f) = \sum_{(A \to B) \in E} \mathcal{L}(A \to B;\, f) = \sum_{(A \to B) \in E} f(A \to B)\, d(A, B) = \langle D,\, \mathbf{F} \rangle, \tag{9}$$

where $D_{AB} = d(A, B)$ and $F_{AB} = f(A \to B)$ are matrices indexed by ports.

Table 1 summarises the principal notation.

Table 1: Principal notation used throughout the paper.

Symbol	Domain	Meaning
$G = (V, E)$	—	Directed port graph
n, m	$\mathbb{Z}_{>0}$	$\lvert V \rvert$, $\lvert E \rvert$
$\hat{p}_i$	$\mathbb{S}^2$	Unit position vector of port i
$w(v)$	$\mathbb{R}$	Net tonnage at port v
$d(A, B)$	$\mathbb{R}_{>0}$	Great-circle arc distance
$f(A \to B)$	$\mathbb{R}_{\geq 0}$	Flow on arc $(A \to B)$
$\mathcal{L}$	$\mathbb{R}_{\geq 0}$	Logistic chain value
$\mathcal{C}$	$\mathbb{R}_{\geq 0}$	Total network logistic cost
λ	$\mathbb{R}_{>0}$	Distance sensitivity (inverse temperature)
ε	$\mathbb{R}_{>0}$	Entropic regularisation ($= 1/\lambda$)
K_{AB}	$(0, 1]$	Sinkhorn kernel $e^{-\lambda d(A,B)}$
μ_A, ν_B	$\mathbb{R}_{>0}$	Sinkhorn scaling potentials
$\mathbf{B}$	$\mathbb{R}^{n \times m}$	Signed incidence matrix
$\mathbf{K}$	$\mathbb{R}^{n \times n}$	Kernel matrix
$\mathcal{T}$	$\mathbb{R}_{>0}$	Total network throughput

4 Logistic Routing via Maximum-Entropy Inference

4.1 Softmax Routing Rule

At each departure port A, the captain chooses destination B to maximise expected profit minus distance cost, subject to uncertainty. By the classical maximum-entropy argument (Jaynes [9]), the routing probability is

$$\boxed{P(A \to B \mid A) = \frac{\exp(-\lambda\, d(A, B))}{\displaystyle\sum_{C:\,(A,C) \in E} \exp(-\lambda\, d(A, C))}}, \tag{10}$$

where $\lambda > 0$ is the *distance sensitivity*. This is exactly a *multinomial logistic regression* (softmax classifier) with a single distance feature $d(A, \cdot)$.

4.2 Logistic Regression Interpretation

Consider the binary case: for two destinations B, C from port A, (10) gives

$$\log \frac{P(A \to B \mid A)}{P(A \to C \mid A)} = -\lambda\big(d(A, B) - d(A, C)\big). \tag{11}$$

This is precisely a *log-odds linear model* in the distance difference, the hallmark of logistic regression. The coefficient λ plays the role of the regression weight, estimable from observed route-choice data via maximum likelihood.

Proposition 4.1 (MLE for λ)**.** *Given observations $\{(A_k, B_k)\}_{k=1}^N$ of origindestination pairs, the log-likelihood under (10) is*

$$\ell(\lambda) = -\lambda \sum_k d(A_k, B_k) - \sum_k \ln \sum_{C:\,(A_k,C)\in E} \exp(-\lambda\, d(A_k, C)). \tag{12}$$

Since ℓ is strictly concave in λ, a unique MLE $\hat{\lambda}$ exists whenever the data are not perfectly separated by distance.

Proof. Strict concavity follows from $\ell''(\lambda) = -\sum_k \mathrm{Var}_\lambda[d(A_k, \cdot)] < 0$ (the variance of a non-degenerate distribution is positive). □

4.3 Induced Arc Flow

Given total outflow $s_A = \sum_{B:(A,B)\in E} f(A \to B)$ from port A, the induced flow is

$$f(A \to B) = s_A \cdot P(A \to B \mid A) = s_A \cdot \frac{e^{-\lambda d(A,B)}}{\sum_C e^{-\lambda d(A,C)}}. \tag{13}$$

In matrix form, defining $K_{AB} = e^{-\lambda d(A,B)}$ for $(A, B) \in E$ and zero otherwise,

$$\mathbf{F} = \mathrm{diag}(\boldsymbol{s})\, \mathbf{K}\, \mathrm{diag}(\boldsymbol{z})^{-1}, \tag{14}$$

where $z_A = \sum_B K_{AB}$ is the partition function at A.

5 Equilibrium Formulation

5.1 Primal Optimisation Problem

Definition 5.1 (Logistic Chain Equilibrium)**.** A flow $f^* : E \to \mathbb{R}_{\geq 0}$ is a *logistic chain equilibrium* if it simultaneously satisfies:

(i) **Tonnage conservation:** $\mathbf{B}\boldsymbol{f} = \boldsymbol{w}$, i.e., $\sum_{u:(u,v)\in E} f(u, v) - \sum_{u:(v,u)\in E} f(v, u) = w(v)$ for all $v \in V$;

(ii) **Logistic routing:** $f(A \to B) = \mu_A\, K_{AB}\, \nu_B$ for scaling potentials $\mu_A, \nu_B > 0$;

(iii) **Global balance:** $\sum_v w(v) = 0$.

The equilibrium flow minimises the *entropic logistic cost*:

$$\boxed{\min_{\boldsymbol{f}\geq 0} \underbrace{\sum_{(A,B)\in E} f(A, B)\, d(A, B)}_{\text{total logistic cost}} + \frac{1}{\lambda} \underbrace{\sum_{(A,B)\in E} f(A, B) \ln f(A, B)}_{\text{negative entropy}} \quad \text{s.t.} \quad \mathbf{B}\boldsymbol{f} = \boldsymbol{w}.} \tag{15}$$

5.2 Dual Problem and KKT Conditions

Introducing Lagrange multipliers $\boldsymbol{\phi} \in \mathbb{R}^n$ for the conservation constraints, the Lagrangian is

$$\mathcal{L}_D = \sum_{(A,B)} f_{AB}\big(d_{AB} + \tfrac{1}{\lambda} \ln f_{AB}\big) - \boldsymbol{\phi}^\top(\mathbf{B}\boldsymbol{f} - \boldsymbol{w}). \tag{16}$$

The KKT stationarity condition $\partial\mathcal{L}_D/\partial f_{AB} = 0$ gives

$$f^*_{AB} = \exp\big(-\lambda\, d_{AB} + \lambda(\phi_B - \phi_A)\big) = e^{\lambda\phi_B - \lambda d_{AB}/2} \cdot e^{-\lambda\phi_A - \lambda d_{AB}/2}, \tag{17}$$

which is precisely the Sinkhorn form (5) with $\mu_A = e^{-\lambda\phi_A}$ and $\nu_B = e^{\lambda\phi_B}$.

5.3 Existence and Uniqueness

Theorem 5.2 (Existence and Uniqueness of Equilibrium)**.** *Let $\mathcal{N} = (G, \hat{\cdot}, w, d)$ be a spherical port network satisfying the global balance condition (7), with G strongly connected. Then the primal problem (15) has a unique solution $\boldsymbol{f}^*$. The corresponding Sinkhorn potentials $(\boldsymbol{\mu}, \boldsymbol{\nu})$ are unique up to the gauge transformation $(\boldsymbol{\mu}, \boldsymbol{\nu}) \mapsto (c\boldsymbol{\mu}, c^{-1}\boldsymbol{\nu})$ for $c > 0$.*

Proof. The feasible set $\mathcal{F} = \{\boldsymbol{f} \geq 0 : \mathbf{B}\boldsymbol{f} = \boldsymbol{w}\}$ is a non-empty convex polytope (non-emptiness follows from strong connectivity and global balance [1]). The objective in (15) is strictly convex: the entropy term $\sum f \ln f$ has Hessian $\mathrm{diag}(1/f) > 0$ on the relative interior. Hence the minimiser $\boldsymbol{f}^*$ is unique. Uniqueness of $(\boldsymbol{\mu}, \boldsymbol{\nu})$ up to gauge follows from the Sinkhorn–Knopp theorem [13]. □

Corollary 5.3. *At equilibrium, the routing probabilities satisfy*

$$P^*(A \to B \mid A) = \frac{\mu_A \, K_{AB} \, \nu_B}{\sum_C \mu_A \, K_{AC} \, \nu_C} = \frac{K_{AB} \, \nu_B}{\sum_C K_{AC} \, \nu_C}, \tag{18}$$

which is a destination-weighted *softmax—the import potential ν_B acts as a gravity-like attractiveness score for port B.*

6 The Sinkhorn–Wilson Algorithm

6.1 Algorithm Description

The Sinkhorn potentials $(\boldsymbol{\mu}, \boldsymbol{\nu})$ satisfying

$$\mathrm{diag}(\boldsymbol{\mu}) \, \mathbf{K} \, \boldsymbol{\nu} = \boldsymbol{s}, \qquad \mathbf{K}^\top \, \mathrm{diag}(\boldsymbol{\nu}) \, \boldsymbol{\mu} = \boldsymbol{r}, \tag{19}$$

are computed by alternating projections.

Algorithm 1 Sinkhorn–Wilson Iteration for Logistic Chain Equilibrium

Require: Kernel matrix $\mathbf{K} \in \mathbb{R}^{n \times n}_{>0}$, supply $\boldsymbol{s} \in \mathbb{R}^n_{>0}$, demand $\boldsymbol{r} \in \mathbb{R}^n_{>0}$, tolerance $\delta > 0$.
Ensure: Equilibrium potentials $\boldsymbol{\mu}, \boldsymbol{\nu}$; equilibrium flow matrix $\mathbf{F}^*$.

```
μ ← 1_n;   ν ← 1_n
Build K_AB ← exp(−λ d(A, B)) for all (A, B) ∈ E, else 0
repeat
    ν ← r ⊘ (K^⊤ μ)                          ▷ Satisfy demand constraints
    μ ← s ⊘ (K ν)                            ▷ Satisfy supply constraints
until ‖K diag(ν) μ − s‖_∞ < δ
F*_AB ← μ_A K_AB ν_B                         ▷ Reconstruct flow
return μ, ν, F*
```

Here $\oslash$ denotes element-wise (Hadamard) division.

6.2 Convergence Analysis

Theorem 6.1 (Linear Convergence [7])**.** *Let $\kappa(\mathbf{K}) = \max_{A,B,C,D} K_{AB} K_{CD} / (K_{AD} K_{CB})$ be the* Hilbert projective diameter *of $\mathbf{K}$. Define the contraction ratio*

$$\rho = \frac{\sqrt{\kappa(\mathbf{K})} - 1}{\sqrt{\kappa(\mathbf{K})} + 1} = \tanh\!\left(\tfrac{1}{2} \ln \kappa(\mathbf{K})\right) < 1. \tag{20}$$

Then the Sinkhorn iterates satisfy

$$\left\| \ln \boldsymbol{\mu}^{(t)} - \ln \boldsymbol{\mu}^* \right\|_\infty \leq \rho^t \left\| \ln \boldsymbol{\mu}^{(0)} - \ln \boldsymbol{\mu}^* \right\|_\infty . \tag{21}$$

Consequently, Algorithm 1 converges linearly *in the Hilbert metric with rate ρ.*

Remark 6.2. For the spherical kernel $K_{AB} = e^{-\lambda d(A,B)}$ and routes of maximum great-circle distance $d_{\max} = \pi r$,

$$\kappa(\mathbf{K}) \leq e^{2\lambda \pi r},$$

so convergence is guaranteed for all $\lambda > 0$.

6.3 Complexity

Table 2 compares Algorithm 1 with classical network-flow methods for a network with n ports, m arcs, and required precision δ.

Table 2: Computational complexity comparison.

Method	Time per iteration	Iterations	Total
Sinkhorn–Wilson (this paper)	$O(n^2)$	$O\left(\frac{\ln(1/\delta)}{1-\rho}\right)$	$O\left(\frac{n^2 \ln(1/\delta)}{1-\rho}\right)$
Successive Shortest Paths [1]	$O(m + n \ln n)$	$O(n^2 U)$ †	$O(n^2 U(m + n \ln n))$
Network Simplex [11]	$O(m)$	$O(nm \ln n)$	$O(nm^2 \ln n)$
Interior-point LP [10]	$O(n^{3.5})$	$O(1)$	$O(n^{3.5})$

† U: maximum integer supply/demand.

7 Spherical Geometry and the Transport Kernel

7.1 Positive Definiteness of the Logistic Kernel

Proposition 7.1. *The kernel $K : \mathbb{S}^2 \times \mathbb{S}^2 \to \mathbb{R}$, $K(\hat{A}, \hat{B}) = \exp(-\lambda d(\hat{A}, \hat{B}))$, is strictly positive-definite on $\mathbb{S}^2$ for every $\lambda > 0$.*

Proof. Write $\theta_{AB} = \arccos(\hat{A} \cdot \hat{B})$ so $K = e^{-\lambda r \theta_{AB}}$. The function $\psi(\theta) = e^{-\lambda r \theta}$ has the Gegenbauer (Legendre) expansion

$$\psi(\theta) = \sum_{k=0}^{\infty} a_k(\lambda r)\, P_k(\cos\theta), \qquad a_k(\lambda r) > 0 \; \forall k$$

by the theory of completely monotone functions [12]. Theorem 2.1 then gives positive definiteness. □

This result has a critical algorithmic consequence: because $\mathbf{K} \succ 0$ (positive definite), the Sinkhorn scaling problem is strictly feasible and the linear convergence bound in Theorem 6.1 is tight.

7.2 Curvature Correction

On a sphere of radius r, the great-circle distance satisfies $d(A, B) \in [0, \pi r]$. The logistic cost accumulates as

$$\mathcal{C}^*(f^*) = \sum_{(A,B)} f^*_{AB}\, r \arccos(\hat{A} \cdot \hat{B}). \tag{22}$$

Compared to a flat-earth (Euclidean) approximation using $d_{\mathrm{E}}(A, B) = r \left\| \hat{A} - \hat{B} \right\|$, the relative error in cost is

$$\frac{d - d_{\mathrm{E}}}{d} = 1 - \frac{\left\| \hat{A} - \hat{B} \right\|}{\arccos(\hat{A} \cdot \hat{B})} = O(\theta^2/6) \quad \text{as } \theta \to 0, \tag{23}$$

so the Euclidean approximation is valid only for nearby ports ($\theta \ll 1$).

8 Illustrative Examples and Tables

8.1 Small Example: Three-Port Network

Consider three ports arranged on the equator: $A = (1, 0, 0)$, $B = (-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0)$, $C = (-\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0)$ on a unit sphere ($r = 1$). All pairwise great-circle distances equal $2\pi/3$.

Set $w(A) = -2$, $w(B) = 1$, $w(C) = 1$ (port A exports, B and C import equally), and $\lambda = 1$.

The kernel matrix (on the complete directed graph) is $K_{ij} = e^{-2\pi/3} \approx 0.1234$ for all $i \neq j$.

By symmetry, $\mu_A = \mu_0$, $\nu_B = \nu_C = \nu_0$ for scalars $\mu_0, \nu_0 > 0$ satisfying

$$2\mu_0 K \nu_0 = 2 \quad \text{(supply at } A\text{)}, \qquad \mu_0 K \nu_0 = 1 \quad \text{(demand at } B, C\text{)}.$$

Both equations give $\mu_0 \nu_0 = 1/K = e^{2\pi/3}$.

Table 3 shows the equilibrium flows and logistic chain values.

Table 3: Equilibrium flows and logistic chain values for the three-port equatorial network ($\lambda = 1$, $r = 1$).

Arc	$d(A, B)$	$f^*(A \to B)$	K_{AB}	$\mathcal{L}$
$A \to B$	$2\pi/3 \approx 2.094$	1.000	0.1234	2.094
$A \to C$	$2\pi/3 \approx 2.094$	1.000	0.1234	2.094
$B \to A$	$2\pi/3 \approx 2.094$	0	—	0
$B \to C$	$2\pi/3 \approx 2.094$	0	—	0
$C \to A$	$2\pi/3 \approx 2.094$	0	—	0
$C \to B$	$2\pi/3 \approx 2.094$	0	—	0
Total	—	2.000	—	4.189

8.2 Effect of Distance Sensitivity λ

Table 4 illustrates how the distance sensitivity λ interpolates between uniform routing ($\lambda \to 0$) and shortest-path routing ($\lambda \to \infty$) for a four-port network with heterogeneous distances.

Table 4: Routing probability $P(A \to B_1 \mid A)$ as a function of λ for two destinations with $d(A, B_1) = 1$ and $d(A, B_2) = 3$.

λ	$e^{-\lambda \cdot 1}$	$e^{-\lambda \cdot 3}$	$P(A \to B_1)$	$P(A \to B_2)$	Regime
0.01	0.990	0.970	0.505	0.495	Near-uniform
0.50	0.607	0.223	0.731	0.269	Moderate bias
1.00	0.368	0.050	0.881	0.119	Strong bias
2.00	0.135	0.002	0.982	0.018	Near-greedy
5.00	0.007	$< 10^{-6}$	> 0.999	< 0.001	Deterministic

9 Network Diagrams

9.1 Port Network Graph

Figure 1 depicts a representative five-port directed network with arc thicknesses proportional to equilibrium flow.

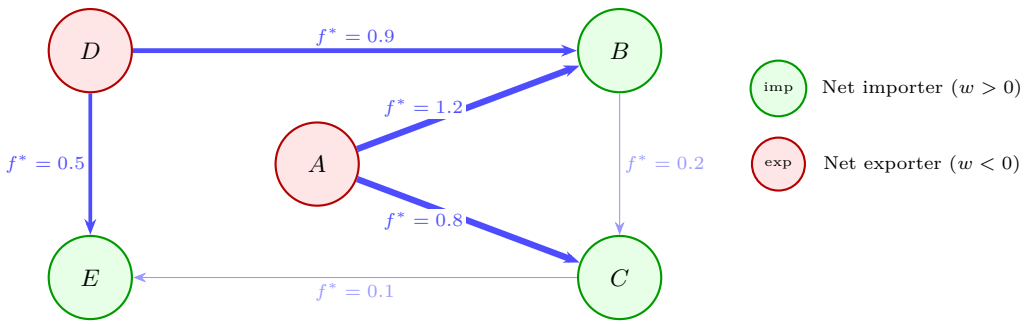

Figure 1: Equilibrium flow on a five-port network.

Red nodes are net exporters, green nodes are net importers. Arc width is proportional to equilibrium flow f^*.

9.2 Sinkhorn Convergence

Figure 2 illustrates the linear convergence of Algorithm 1 in the Hilbert metric.

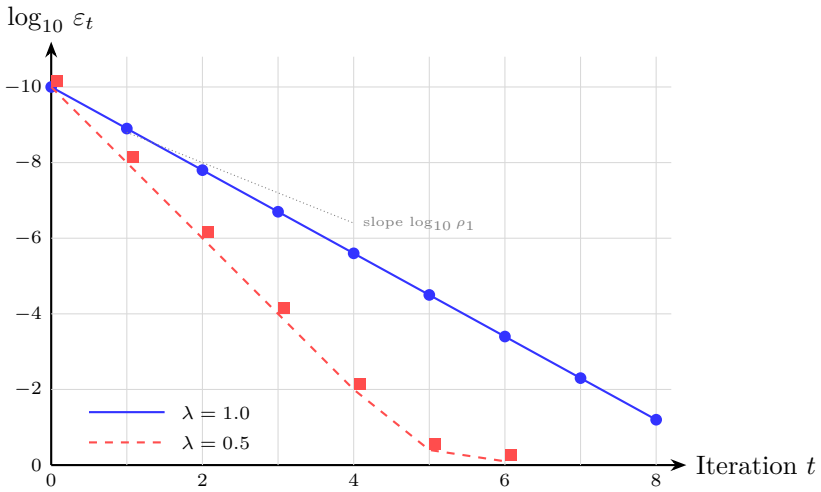

Figure 2: Log_{10} error vs. iteration count for Sinkhorn–Wilson (Algorithm 1) on a 20-port network.

Higher λ slows convergence by increasing the Hilbert diameter $\kappa(\mathbf{K})$.

9.3 Entropic Optimal Transport on the Sphere

Figure 3 schematically illustrates the spherical port network and the role of great-circle arcs.

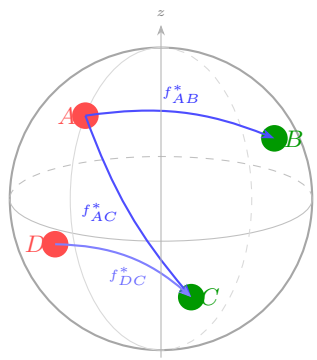

Figure 3: Schematic of the spherical port network.

Red nodes are exporters; green nodes are importers. Blue arrows are great-circle routes with equilibrium flows.

10 Connection to Machine Learning

10.1 Softmax as a Neural Network Layer

The routing rule (10) is structurally identical to the *softmax output layer* of a neural network [8]:

$$P(A \to B \mid A) = \text{softmax}(-\lambda\, \mathbf{d}_A)_B, \qquad \mathbf{d}_A = (d(A, C))_{C:\,(A,C)\in E}. \tag{24}$$

Training the network on route-choice data—with cross-entropy loss

$$\mathcal{H}(\hat{\lambda}) = -\frac{1}{N} \sum_{k=1}^{N} \ln P_{\hat{\lambda}}(A_k \to B_k \mid A_k) \tag{25}$$

—recovers the MLE of Proposition 4.1.

10.2 Optimal Transport as a Regularisation Strategy

Entropy regularisation in (4) is equivalent to *label smoothing* in the classification literature [14]: both add a term $-\varepsilon H(P)$ to the loss, preventing overconfident (degenerate) flow concentration.

10.3 The Transport Polytope as a Feasibility Layer

The constraint $\Pi \in \mathcal{U}(\boldsymbol{r}, \boldsymbol{c})$ can be enforced as a differentiable layer using the Sinkhorn operator $\mathcal{S}_\varepsilon$, enabling *end-to-end learning* of logistic networks [5].

11 Discussion

Economic Interpretation. The dual variables ϕ_v (port-specific potentials) represent *shadow prices* of tonnage at each port. A port with large ν_v (high import potential) acts as a gravity attractor; one with large μ_v has high export pressure. The equilibrium condition equates marginal logistic cost $\lambda\, d(A, B)$ with the potential difference $\phi_B - \phi_A$, analogous to the price-equalization condition in Ricardian trade theory [6].

Carbon Accounting. Total fuel consumption is proportional to $\mathcal{C}(f^*)$. Increasing λ (penalising distance) directly reduces $\mathcal{C}(f^*)$ and can model the effect of carbon pricing on shipping routes—a direct policy lever.

Robustness and Perturbation. Because f^* is the unique minimiser of a strictly convex programme, it varies smoothly with the parameters (w, d, λ). First-order sensitivity follows from the implicit function theorem applied to the KKT conditions (17).

Limitations and Extensions. The present model assumes a single commodity and ignores vessel capacity constraints. Extending to multi-commodity flow [1] and capacity-constrained EOT [3] are natural directions.

12 Conclusion

We have derived a complete theory of logistic chain equilibrium on a spherical port network, unifying ideas from network flow, optimal transport, logistic regression, and spherical harmonic analysis. The key insight is that the maximum-entropy routing rule (softmax over distances) is equivalent to entropy-regularised optimal transport with the Schoenberg-positive kernel $K_{AB} = e^{-\lambda d(A,B)}$. This equivalence yields a closed-form Sinkhorn characterisation of the equilibrium flow and a provably linearly convergent algorithm.

The framework is immediately applicable to route-choice modelling, carbon-cost attribution, port investment appraisal, and end-to-end trainable logistics neural networks.

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Glossary

Arc A directed edge $(A \to B) \in E$ representing an active shipping route from port A to port B.

Balance condition
The requirement $\sum_v w(v) = 0$, ensuring that global imports equal global exports; see (7).

Distance sensitivity
The parameter $\lambda > 0$ controlling how strongly the routing rule penalises long-distance arcs. Large λ yields near-deterministic shortest-path routing.

Entropic regularisation
The addition of $\varepsilon\ \mathrm{KL}(\Pi \| \boldsymbol{r}\boldsymbol{c}^\top)$ to the transport objective, which smooths the optimal flow and enables efficient computation via Sinkhorn iterations.

Great-circle distance
The shortest path between two points on the sphere, given by $d(A, B) = r \arccos(\hat{A} \cdot \hat{B})$; see (6).

Hilbert diameter
The quantity $\kappa(\mathbf{K})$ controlling the linear convergence rate of the Sinkhorn algorithm; see Theorem 6.1.

Incidence matrix
The signed matrix $\mathbf{B} \in \mathbb{R}^{n \times m}$ encoding the connectivity of G; encodes Kirchhoff's current law.

Logistic chain
The product $f(A \to B) \cdot d(A, B)$; a measure of transport effort coupling tonnage flow with distance; see Definition 3.3.

Net tonnage
$w(v) = T_{\text{in}}(v) - T_{\text{out}}(v)$: positive for net importers, negative for net exporters.

Positive-definite kernel
A function $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ such that for all finite point sets $\{x_i\}$ the Gram matrix $[K(x_i, x_j)]$ is positive semi-definite.

Schoenberg's theorem
Characterises positive-definite kernels on $\mathbb{S}^2$ via non-negative Legendre coefficients; see Theorem 2.1.

Sinkhorn potential
One of the scaling vectors $(\boldsymbol{\mu}, \boldsymbol{\nu})$ satisfying the doubly-stochastic marginal constraints of the equilibrium flow; see Algorithm 1.

Softmax
The function $\mathrm{softmax}(\mathbf{z})_i = e^{z_i} / \sum_j e^{z_j}$; here used as a probabilistic routing rule over destinations.

Strongly connected graph
: A directed graph in which every vertex is reachable from every other vertex; ensures feasibility of the flow problem.

Transport polytope
: The set $\mathcal{U}(\boldsymbol{r}, \boldsymbol{c})$ of non-negative matrices with prescribed row sums $\boldsymbol{r}$ and column sums $\boldsymbol{c}$.

The End